The Riemann Hypothesis
A Synthesis of the Beurling–Nyman Approach

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Abstract

We synthesize the Beurling–Nyman approach to the Riemann Hypothesis. The criterion: RH is equivalent to $d_N \to 0$, where d_N is the $L^2(0,1)$ distance from $1_{(0,1)}$ to the span of the BN functions $\{\rho_{1/k}\}_{k=2}^{N+1}$. This body has established: the Gram matrix is approximately Toeplitz with symbol $\sigma(t) = |\zeta(1/2+it)|^2/(1/4+t^2)$; the Killip–Simon theorem gives Toeplitz convergence unconditionally; the gap between Toeplitz and BN is exactly what RH controls; the explicit formula makes d_N^2 equal to the prime counting error; the Hadamard product identifies the wall as thick zero factors; the density spectrum quantifies the permanent residual; and the pair correlation connects the BN angle sum to GUE statistics. Three formulations of the wall are identified, all equivalent to each other and to RH. The wall stands.

1 The Criterion

Definition 1 (BN functions and distance)**.** For $k \geq 2$: $\rho_{1/k}(x) = \{1/(kx)\} - (1/k)\{1/x\}$, where $\{y\}$ is the fractional part. The span: $\mathcal{V}_N = \operatorname{span}\{\rho_{1/2}, \ldots, \rho_{1/(N+1)}\} \subset L^2(0,1)$. The BN distance: $d_N^2 = \inf_{f \in \mathcal{V}_N} \|1_{(0,1)} - f\|_{L^2}^2$.

Theorem 2 (Nyman–Beurling–Báez-Duarte)**.** *$d_N \to 0$ as $N \to \infty$ if and only if the Riemann Hypothesis holds.*

The BN flow is the greedy Gram–Schmidt process on $\{\rho_{1/k}\}$ toward $1_{(0,1)}$: at each step it adds the BN function $\rho_{1/N}$ and orthogonalizes. The angles θ_N satisfy $\cos^2\theta_N = 1 - d_N^2/d_{N-1}^2$, and $d_N^2 = d_1^2 \prod_{k=2}^{N} \sin^2\theta_k$. RH is equivalent to $\sum \cos^2\theta_N = \infty$.

2 The Gram Matrix

Theorem 3 (Gram matrix structure [1, 2, 3])**.** *The BN Gram matrix G_N with (k,l)-entry $\langle \rho_{1/k}, \rho_{1/l} \rangle$ satisfies:*

$$G_N \approx D\, T_N(\sigma)\, D,$$

where $D = \operatorname{diag}(k^{-1/2})_{k=2}^{N+1}$, $T_N(\sigma)$ is the $N \times N$ Toeplitz matrix with symbol $\sigma(t) = |\zeta(\frac{1}{2} + it)|^2/(\frac{1}{4} + t^2)$, and the correction $\delta G_N = G_N - DT_N(\sigma)D$ satisfies: under RH, $\|\delta G_N\|_{\mathrm{op}} = O((\log N)^2/N)$; under $\neg$RH, $\|\delta G_N\|_{\mathrm{op}} = \Omega(N^{-(2\sigma^*-1)+\varepsilon})$ where $\sigma^* = \sup \operatorname{Re}(\rho)$.

Theorem 4 (Killip–Simon–Widom [3, 4]). *Since* $\int_{-\infty}^{\infty} \log \sigma(t)/(1+t^2)\,dt = -\infty$ *(unconditionally), the Szegő–Widom theorem gives* $\det T_N(\sigma) \to 0$ *super-exponentially. The Killip–Simon theorem applied to* $T_N(\sigma)$*: the Verblunsky coefficients* α_N^{Toep} *of the exact Toeplitz model satisfy* $\sum |\alpha_N^{\mathrm{Toep}}|^2 = \infty$ *unconditionally. The gap:* $\alpha_N^{\mathrm{BN}} = \alpha_N^{\mathrm{Toep}} + \delta_N$*, and* $\delta_N \in \ell^2$ *if and only if RH.*

3 The Prime Structure

Theorem 5 (Prime dominance [5, 6]). *For prime p:* $\cos^2 \theta_{p-1} \geq c(\log p)^2/p$*. For composite n:* $\cos^2 \theta_{n-1} \ll \cos^2 \theta_{p-1}$ *for the nearest prime* $p \leq n$*. Therefore:*

$$S_N = \sum_{k=1}^{N} \cos^2 \theta_k \geq c \sum_{p \leq N} \frac{(\log p)^2}{p} \geq c' \frac{\log N}{\log \log N}.$$

Numerically: $S_{150} = 3.405$*, rate* $S_N \sim 1.1 \log N / \log \log N$*. Under RH:* $d_N^2 = O(N^{-c/\log \log N})$*.*

4 The Explicit Formula

Theorem 6 (BN distance as prime error [7]). *Let* $E(x) = \psi(x) - x$ *be the prime counting error. Then:*

$$d_N^2 \asymp \int_N^{2N} \frac{|E(e^u)|^2}{e^{2u}}\,du.$$

The BN distance is the L^2 *norm of the normalized prime error on the dyadic interval* $[N, 2N]$*. Under RH:* $E(x) = O(\sqrt{x}\log^2 x)$*, consistent with* $d_N^2 = O(N^{-c/\log \log N})$*.*

5 The Hadamard Wall

Theorem 7 (Thick and thin zeros [8]). *Each nontrivial zero* $\rho = \sigma + i\gamma$ *of* ζ *contributes a factor* $F_\rho(t) = (\sigma + i\gamma)/(\sigma - 1/2 + i(\gamma - t))$ *to* $1/\zeta$ *on the critical line.*

- *Thin zero* ($\sigma = 1/2$)*:* $|F_\rho(t)| \to \infty$ *as* $t \to \gamma$*; the BN flow can cancel this factor asymptotically.*
- *Thick zero* ($\sigma > 1/2$)*:* $|F_\rho(t)| \geq (\sigma^2 + \gamma^2)^{1/2}/(\sigma - 1/2) > 1$ *uniformly; the factor cannot be canceled.*

The permanent residual: $d_\infty^2 = \sum_{\mathrm{Re}(\rho) > 1/2} (\mathrm{Re}(\rho) - 1/2)^2/|\rho|^2 = 0$ *iff RH.*

6 Three Forms of the Wall

Theorem 8 (The wall in three languages). *The following are equivalent, each a formulation of what must be proved:*

1. ***Arithmetic***: $\sum \cos^2\theta_N = \infty$ *unconditionally. Equivalently:* $d_N^2 = O(N^{-\delta})$ *for some* $\delta > 0$.

2. ***Analytic***: *the non-Toeplitz correction* $\delta_N \in \ell^2$ *without assuming RH; or the oscillatory integral* $|\langle r_{N-1}, \rho_{1/N}\rangle| \geq c d_{N-1}/(N^{1/2}(\log N)^{1/2})$ *unconditionally.*

3. ***Spectral***: *the density spectrum* $\Lambda(1/2^+) = 0$*; no zero of* ζ *is thick; every Hadamard factor is thin.*

Each is equivalent to RH. There is no shortcut: proving any one unconditionally proves all three and proves RH.

7 The Pair Correlation

Theorem 9 (GUE and the flow [9])**.** *Under RH and the Montgomery conjecture (GUE statistics for zeros):*

$$\cos^2\theta_N \geq \frac{c}{\log N}, \quad S_N \geq c\frac{\log N}{\log\log N}, \quad d_N^2 \to 0.$$

Montgomery (1973): RH $\Rightarrow$ GUE (for $|\alpha| < 1$*). The BN chain: GUE $\Rightarrow$ RH. Conjectured: RH $\iff$ GUE. If true: a proof of GUE from first principles would prove RH, and conversely. The pair correlation is not a new approach; it is the wall in the language of random matrices.*

8 State of the Art

Theorem 10 (What is proved)**.** *The following hold unconditionally:*

1. $d_1^2 = 1 - \log 2$ *(exact).*

2. $\int \log\sigma = -\infty$*: the continuum prediction error is 0.*

3. $S_N \geq c(\log N)^{1/3}/(\log\log N)^{1/3}$ *(Vinogradov–Korobov rate, Theorem 3).*

4. $d_N^2 = O(\exp(-c(\log N)^{1/3}/(\log\log N)^{1/3}))$.

5. *The Gram matrix* G_N *is computable and satisfies* $G_N \approx DT_N(\sigma)D$ *with an explicit, bounded correction.*

6. *The permanent residual* d_∞^2*, if positive, is given by the formula of Theorem 7, convergent under any zero-density hypothesis.*

The following are equivalent to RH (not proved):

1. $d_N^2 \to 0$.

2. $S_N \to \infty$ *at rate* $\geq c\log N/\log\log N$.

3. *The Verblunsky correction* $\delta_N \in \ell^2$.

4. *The oscillatory integral has the lower bound in Theorem 8(2).*

5. *No zero of ζ is thick (Theorem 7).*

Remark 11 (The wall). Every approach reaches the same obstruction: the non-Toeplitz correction, the oscillatory lower bound, the permanent residual, the thick zero. These are four names for the same wall. The wall is not ignorance; it is precision. We know exactly what needs to be proved. We do not know how to prove it.

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